GGG–J–51e
March 29, 1967
SUPERSEDING
Int. Fed. Spec. GGG–J–0051d(GSA–FSS)
September 8, 1964 and
Fed. Spec. GGG–J–51c
November 26, 1957 (In part) (See 6.4)

FEDERAL SPECIFICATION

JACKS, MECHANICAL, HAND OPERATED (SCREW; SCISSORS; REEL; PLANER; AND RACK BAR)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers mechanical jacks for heavy duty use in handling heavy machinery, automotive equipment, and construction and maintenance equipment. (For detail information see 6.1.)

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the titles of the specifications, but include only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. Jacks covered by this specification shall be of the following types, classes, and styles, as specified (see 6.2):

Type I.—Screw, hand.
- Class 1.—Standard speed.
  - Style A.—Upright or inverted, with foot lift.
  - Style B.—Journal or inverted.
- Class 2.—High-speed, lever- or governor-lowering control.
- Class 3.—Bell bottom.
- Class 4.—Geared screw (for auto or truck).

Type II.—Scissors, hand.

Type III.—Reel, hand, (cable, wire, and wire rope).
- Style A.—Rack bar.
- Style B.—Screw.

Type V.—Planer (leveling).

Type VI.—Rack bar (ratchet-lever lowering).
- Class 1.—Solid base with foot lift.
  - Style A.—Single socket.
  - Style B.—Double socket.
- Class 2.—Hinged base, double socket.
  - Style A.—With foot lift.
  - Style B.—Pole pulling.

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2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

MM-L-736—Lumber and Timber; Hardwood.

RR-C-271—Chains and Attachments, Welded, Weldless, and Roller Chain.

Federal Standards:

Fed. Std. No. 66—Steel: Chemical Composition and Hardenability.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools; Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

National Bureau of Standards Handbook:

H28—Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C., 20402.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of jacks which are otherwise in accordance with the requirements of this specification.

3.2 Design and construction. The jacks shall be of the latest design, convenient to operate, and entirely adequate for the purpose intended. All parts subject to wear and normally replaceable shall be readily accessible for adjustment or repair. The load shall be lowered or released only when, and as, controlled by the operator. Jacks shall be of safe construction for their rated capacity, and shall be capable of long life under ordinary service conditions.

3.2.1 Capacities. The capacities stated for the various types, classes, and styles of jacks shall be for the head lift (see table XIV). The capacities of the claw or foot lift shall be not less than one-half of the head lift.

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3.2.2 Castings. All castings shall be of uniform quality, free of blowholes, porosity, hard spots, shrinkage defects, cracks, or other injurious defects. Strength and other essential physical properties of the castings shall be adequate throughout for the purpose intended.

3.2.3 Welding, brazing, and leading. Welding, brazing, and leading processes shall be performed in accordance with good commercial practice. In no event shall such processes as peening or plugging be used on castings or forgings for reclaiming any parts of the jack without authorization of the procuring agency.

3.2.4 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with an adequate means for preventing loss of proper tightness and adjustment.

3.2.5 Lubrication. All bearings requiring periodic lubrication shall be provided with accessible means of lubrication. Each jack shall be adequately lubricated.

3.2.6 Threads. All threaded parts shall be in accordance with the applicable requirements of H28.

3.2.7 Bushings. Bushings shall be of steel, bronze, or other suitable metal.

3.2.8 Surface finish. All castings, forgings, molded, or welded parts shall be thoroughly cleaned and shall be free of sand, dirt, fins, sprues, scale, flux, or other harmful extraneous materials. All external surfaces and edges shall be free of burrs, fins, and sharp edges.

3.2.9 Top surface. The cap, or top surface of all jacks, except type I, class 4, types II and III, shall be corrugated or otherwise suitably roughened or designed to prevent slipping of the load.

3.3 Safety devices. Each jack shall incorporate a positive stop to prevent overtravel. If this is impractical, the jack shall be equipped with warning grooves or stop indicator to indicate the travel limit of safe travel.

3.4 Lever. A suitable operating lever shall be furnished with each jack, except type V. The lever shall be of such length and strength that the rated load may be raised without injury to any part of jack or lever. Raising or lowering of the jack shall be accomplished without the aid of special tools, unless incorporated in the removable handle or lever.

3.5 Identification marking. Jacks shall be marked in a plain and permanent manner with the manufacturer's name, or with a trademark of such known character that the source of manufacture may be readily determined. Jacks shall also be marked with the rated capacity of the jack in tons and the jack model, except that type I, class 3, type II, and type V jacks are not required to have the model marked thereon. Unless otherwise specified (see 6.2), decals may be used for identification marking on type II. For protection, decals shall be located on a protective area of the jack when practicable. When specified (see 6.2), decals shall have an acrylic coating, or a coating of other equally suitable material.

3.6 Equipment. Such additional equipment as required may be specified (see 6.2), and shall be furnished with the jacks.

3.7 Finish. Unless otherwise specified (see 6.2), exposed surfaces of the jack shall be coated or finished according to the manufacturer's standard practice.

3.8 Repair parts. The successful bidders supplying jacks covered by this specification, shall make available, and furnish, repair parts when requested. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances in order that such parts may be completely interchangeable, replaced or adjusted without requiring modification.

3.9 Type I, screw, hand.

3.9.1 Type I, class 1, standard speed, style A, upright or inverted, with foot lift. Type I, class 1, style A jacks shall consist essentially of a base, gearing mechanism, suitable shaft between the pinion and ratchet, sleeve, standard, and screw with engaging threads in the standard lining cap or nut, ratchet, and detachable lever. Drop-type carrying handles shall be provided. Jacks shall be ball or roller bearing and provided with a lifting sleeve cast in one piece or a steel standard having a solid or removable top and equipped with a claw or foot lift which may be plain or roughened, according to the manufactur-

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er's standard practice. The base shall be of suitable size to prevent the jack from tilting when load is placed on the claw.

3.9.1.1 Gearing mechanism. The gearing mechanism shall consist of either a gear and pinion made of steel, or a worm and worm gear, the worm being of steel and the worm gear of bronze. All gearing shall have machine-cut teeth. The pinion may be forged integrally with the part of the shaft between the pinion and the ratchet, or may be secured to the shaft in any other equally satisfactory manner. The gearing mechanism shall be maintained in proper mesh with a minimum of lost motion.

3.9.1.2 Ratchet mechanism. The ratchet shall be of steel and the ratchet holder of malleable iron or steel. The pawl or detent shall be of steel properly heat treated and shall engage the teeth of the ratchet gear. Means shall be provided to change the operating direction of the jack.

3.9.1.3 Main bearing on which the screw rotates shall consist of two hardened-steel plates, separated by balls or rollers made of hardened alloy steel.

3.9.1.4 Base, standard, and sleeve. The sleeve shall be of steel or malleable iron. The standard and the base shall be of malleable iron or steel. A bushing or nut of alloy bronze or other suitable metal threaded internally to engage with the screw shall be fitted inside the standard.

3.9.1.5 Screw. The screw shall be of steel and the threads cut smoothly. The threaded portion of the screw shall engage with the threads in the lining or nut in the standard. The other end shall project through the bearing plates. The end of the screw which passes through the bearing plates shall rotate freely.

3.9.1.6 Type I, class 1, style A jacks shall conform to table I for the capacity and height specified (see 6.2), and be similar to a or b of figure 1.

Table I. Type I, class 1, style A. screw, hand jacks, standard speed, upright, or inverted, with foot lift.

Rated capacity	Closed height (maximum)	Rise (minimum)	Size of base (minimum)	Weight of jack (maximum)	Head (cap) diameter (minimum)
Tons	Inches	Inches	Square inches	Pounds	Inches
15	22	10	55	182	3¾
25	22	9	60	170	4¼
25	26	10	60	150	4¼
35	27	10	71	210	5⅛
50	24	14	108	255	5¼
50	27	10	93	300	6⅛
75	27	12	125	400	6⅛

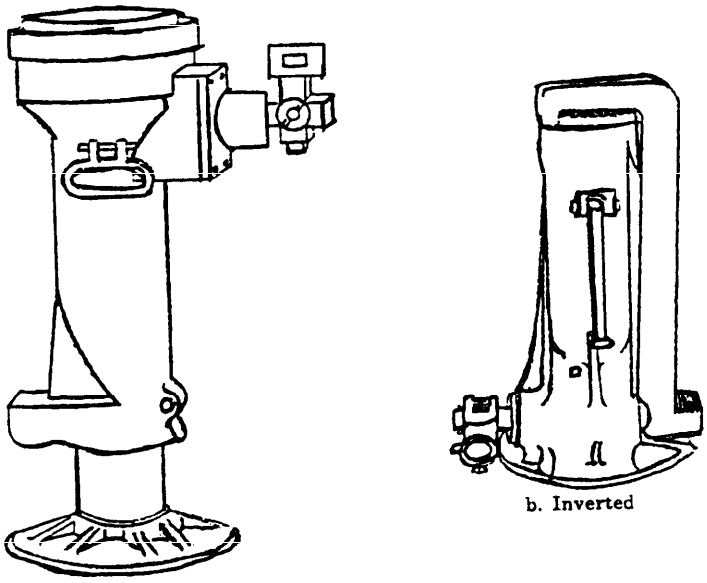
3.9.2 Type I, class 1, standard speed, style B, journal or inverted. Type I, class 1, style B jacks shall consist essentially of a gear and pinion, suitable shaft between pinion and ratchet; casing, ram, cap, and screw with engaging threads in the ram; ratchet and ratchet holder, and detachable lever.

3.9.2.1 Gear and pinion. The gear and pinion shall be of steel and shall have machine-cut teeth. The pinion may be forged integrally with the shaft between the pinion and the ratchet, or secured to the shaft in any other equally satisfactory manner. The design shall be such that the gear and pinion will be in proper mesh with a minimum of lost motion.

3.9.2.2 Ratchet mechanism. The ratchet shall be of steel and the ratchet holder of malleable iron or steel. The pawl or detent shall be of steel properly heat treated, and shall engage the teeth of the ratchet gear. Means shall be provided to change the direction of operation of the jack.

3.9.2.3 Main bearing. The main bearing on which the screw rotates shall consist of two hardened-steel plates separated by balls or rollers made of hardened alloy steel.

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a. Upright

b. Inverted

FIGURE 1.—Type I, class 1, style A, screw hand jacks, standard speed, upright or inverted, with foot lift.

3.9.2.4 Casing and ram. The casing of the jack shall be of malleable iron, steel, or aluminum alloy, and shall be equipped with an integral or drop-type carrying handle suitable for moving the jack. The ram shall be of steel or seamless-steel tubing and fitted with a nut or bushing made of alloy bronze or other suitable metal threaded to engage the threads of the screw. The ram shall be designed to prevent rotation under load, and shall be fitted with a forged or cast-steel top.

3.9.2.5 Screw. The screw shall be of steel and the threads cut smoothly. One end of the screw shall be threaded into the ram, and the other end securely fastened to the gear.

3.9.2.6 Type I, class 1, style B jacks shall conform to table II for the capacity and height specified (see 6.2), and be similar to figure 2.

Table II. Type I, class 1, style B, screw, hand jacks, standard speed, journal or inverted.

Rated capacity	Closed height (maximum)	Rise (minimum)	Weight of jack (maximum)	Size of base (minimum)	Head (cap) diameter (minimum)	Lever length (minimum)
Tons	Inches	Inches	Pounds	Square inches	Inches	Inches
15	7	2½	22	17.7	2¾	28
15	10	5	40	17.7	2⅝	25
15	14¾	9	50	28	3⅛	28
25	10	5	55	19.6	3	35
25	15	9	40	44.2	3⅛	35
35	10¼	5	60	22	3⅛	35
50	10	4	120	37.1	4½	50
50	16½	7	152	37.1	4½	50

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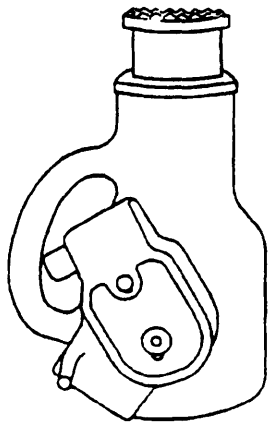


FIGURE 2.—Type I, class 1, style B, screw, hand jack, standard speed, journal or inverted.

3.9.3 Type I, class 2, high speed, lever- or governor-lowering control.

3.9.3.1 General design and component parts. Type I, class 2 jacks shall be of the regular (operating mechanism in the head) or the inverted (operating mechanism in the bottom) lever- or governor-control self-lowering jacks having a main screw, the thread of which shall be such as to permit the jack to be lowered gradually under any load within the rated capacity (and test capacities), by means of a lever-control self-lowering advice. These jacks shall conform to class 1 jacks, except that the following additional essential features shall be included: Gearing mechanism, thrust bearings, intermediate gearing, lever-control lowering device, self-locking emergency device, automatic self-lowering device and control, and unless otherwise specified, shall be equipped with wheels or suitable carriage to facilitate their being moved as required.

3.9.3.2 Gearing mechanism. The spur gears and worm shall be of steel. The worm gear shall be of bronze. All gearing shall have machine-cut teeth. The design shall be such that proper mesh shall be effected with a minimum of lost motion.

3.9.3.3 Bearings and bushings. Thrust bearings shall be of ball or roller type. Bushings shall be bronze or other suitable bushing material.

3.9.3.4 Intermediate gearing. Intermediate gearing shall be steel, machine-cut, heat treated and finished, and shall be divided by a clutch arrangement of bronze or other suitable metal and steel, including drop-forged, spring-actuated engaging and disengaging pawls of suitable design, thus permitting the jack to rise while the lowering gear is in locked position.

3.9.3.5 Lever- or governor-lowering control. The self-lowering thumb- or governor-control lowering lever shall be of malleable iron or steel, engaging the cantilever brake lever through which the operator may exert normal pressure (leverage) at will, against a brake wheel at the terminus of the gear train for starting and stopping the lowering mechanism of the jack. The brake levers shall, by suitable spring or screw pressure, be automatically locked securely against the brake wheel except when the jack is being lowered. A lock shall be provided such as a threaded rod extending through a gear housing designed and constructed to securely hold the load, and shall be released by means of a simple mechanical device which shall permit the jack to lower the load of its own accord, at such lowering speeds as may be desired by the operator.

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3.9.3.6 Carrying handles. Two or more drop-type metal carrying handles shall be provided for lifting and carrying the jack.

3.9.3.7 Lubrication. The operating mechanism shall run in oil or grease.

3.9.3.8 Type I, class 2 jacks shall conform to table III for the capacity and height specified (see 6.2), and be similar to figure 3.

Table III. Type I, class 2, screw, hand jacks, high-speed, lever- or governor-lowering control.

Rated capacity	Closed height (maximum)	Rise (minimum)	Weight of jack (maximum)	Size of base (minimum)	Head (cap) diameter (minimum)
Tons	Inches	Inches	Pounds	Square inches	Inches
25	28	15	175	113	4
35	28	15	210	85	4
50	26	13	340	100	5
50	30	15½	350	100	5
50	36	23	350	100	5½
75	26	12	450	130	6⅛
100	26	12	500	150	7⅛

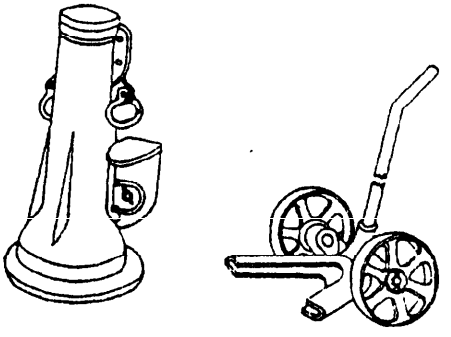


FIGURE 3.—Type I, class 2, screw, hand jack, high-speed, lever- or governor-lowering control.

3.9.4 Type I, class 3, bell bottom.

3.9.4.1 Component parts. Type I, class 3 jacks shall be the ordinary type of screw jack consisting essentially of a barrel, screw, head, cap, and lever.

3.9.4.2 Barrel. The barrel shall be of cast iron, malleable iron, or steel and shall be threaded internally to engage the threads of the screw. It shall be the form known as a bell base or equivalent, the barrel widening out to give a large bearing surface at the base. The bottom of the barrel shall be finished or cast perpendicular to the longitudinal axis of the screw. A handle, forming an integral part of the barrel, shall be provided for carrying the jack. The barrel shall have a peep hole to permit the operator to check against overtravel of the jack.

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3.9.4.3 Screw. The screw shall be of steel, in one piece, and threaded externally to engage the threads in the barrel. The upper end of the screw shall extend up through the barrel and a head and cap shall be secured at the top.

3.9.4.4 Head and cap. The head shall be permanently secured to the screw below the cap (or may be integral with the screw), and contain holes of sufficient size for inserting an appropriate lever to turn the head and screw to raise the load. The head holes shall be of the four-way type. The cap shall be so secured to the screw that it shall not be removable, and shall rotate freely with slight lateral movement. Plain friction-type caps or ball-bearing caps shall be used. Friction-type caps shall be stress-relieved steel, malleable iron, or a high-strength cast iron of at least 40,000 p.s.i. tensile strength. Ball-bearing type caps shall be of malleable iron or steel. A lateral movement of not more than ⅛ inch in the ball-bearing caps will be permitted. Ball bearings shall be spherical, entirely covered by the cap, and fully protected from dirt and dust.

3.9.4.5 Type I, class 3 jacks shall conform to table IV for the capacity specified (see 6.2), and be similar to figure 4.

Table IV. Type I, class 3, screw, hand jacks, bell-bottom.

Rated capacity	Closed height (maximum)	Safe rise of screw (minimum)	Diameter of screw (minimum)	Weight of jack (maximum)	Size of cap (diameter) (minimum)	Size of base (minimum)	Length of lever (minimum)
Tons	Inches	Inches	Inches	Pounds	Inches	Square inches	Inches
5	9¾	3½	1¼	11	2⅞	15	18–25
8	12¼	5	1½	15	2⅞	21	20–25
12	14¼	7	1¾	23	3⅛	30	24–37
15	16¼	9	2	32	3⅛	35	24–37

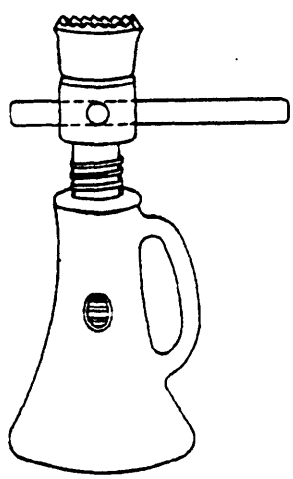


FIGURE 4.—Type I, class 3, screw, hand jack, bell bottom.

3.9.5 Type I, class 4, geared screw (for auto or truck). Type I, class 4 jacks shall be designed for automobiles and motor trucks. The base of each jack shall be sufficiently large in area to insure stability when the jack is in use under service conditions. Operating handles longer than 24 inches shall be

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made to telescope or fold when being carried. The jack shall operate easily and efficiently by means of the operating handle. The screw shall show no signs of failure when tested with the maximum screw extension under test load requirements.

3.9.5.1 The jacks shall have a mechanism which includes reduction gears to reduce the power application to a minimum, and shall be fitted with ball or roller bearings or a friction disk to reduce friction loss under load. The jack housing shall be of malleable iron or steel. A flip type of cap may be furnished at the option of the contractor. Each jack shall be furnished with a suitable operating handle.

3.9.5.2 Type I, class 4 jacks shall conform to table V for the capacity specified (see 6.2), and be similar to figure 5.

Table V. Type I, class 4, screw, hand jacks, geared screw (for auto or truck).

Rated capacity	Closed height (maximum)	Rise (minimum)	Size of base (minimum)	Size of cap (head) (minimum)	Lever length (minimum)	Diameter of screw (minimum)
Tons	Inches	Inches	Square inches	Square inches	Inches	Inches
1½ (auto)	7 5/16	8½	17	1.62	45	5/8
2 (auto)	7¼	9	21	1.25	40	¾
3 (auto or truck)	9¾	6	25	2.25	23	1
5 (auto or truck)	8⅜	10	35	1.75	24	1

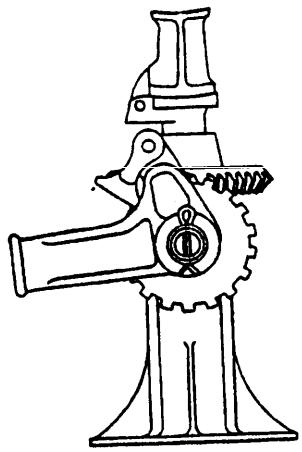


FIGURE 5.—Type I, class 4, screw, hand jack, geared screw (for auto or truck).

3.10 Type II, scissors, hand. Type II jack linkages shall be of steel so formed and fabricated as to raise and sustain the rated and test capacity loads rigidly, without twisting, bending or permanent set, and shall be secured by suitable steel reinforcements, rivets, and bearing surfaces. The horizontal screw may be provided with a one-way thread or with a right- and left-hand thread. When the horizontal screw is furnished with a one-way thread it shall be provided with a suitable thrust bearing to permit easy operation of the jack while under load.

3.10.1 Operating handle. The handle shall be such as to readily permit placing and operating the jack at any desired location under the motor vehicle.

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3.10.2 Type II jacks shall conform to table VI for the capacity specified (see 6.2), and be similar to figure 6.

Table VI. Type II, scissors, hand jacks.

Rated capacity	Closed height (maximum)	Rise (minimum)	Size of base (minimum)	Size of cap (head) (minimum)	Lever length (minimum)	Diameter of screw (minimum)
Tons	Inches	Inches	Square inches	Square inches	Inches	Inches
1½ (auto) -----	5	9¼	20	7½	45	⅝
2 (auto) -------	4½	10	20	9	45	¾

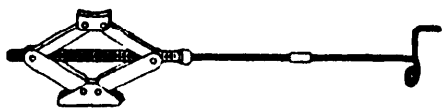


FIGURE 6.—Type II, scissors, hand jack.

3.11 Type III, reel, hand (cable, wire and wire rope). Type III jacks shall be of the rack bar (style A), or screw (style B) type for handling reels of cable, wire, or wire rope. The top of the lifting cap shall be U shaped to cradle the steel bar or axle. A minimum of two hooks shall project from the side of the style A jack body to provide extra lifting levels. The side hooks shall have a 2-inch minimum spindle capacity. Jack construction shall be of such design that jacks will not tip over when maximum diameter reels are handled.

3.11.1 Axle. When specified (see 6.2), a steel axle shall be furnished with each pair of reel jacks for supporting heavy reels. The axle shall be 2-inch nominal diameter, at least 6 feet long, and chamfered in both ends. Axle shall be of hot-rolled steel conforming to AISI 4140H of Fed. Std. No. 66, and heat treated to a Rockwell C hardness between 29 aand 33.

3.11.2 Style A, rack bar. Rack bar jacks shall consist essentially of a cap, base and frame, rack bar, retaining and lifting pawls, and a wood base. Jacks of 5-ton capacity shall be of the round, single-socket, ratchet-action type, and provided with a lever not less than 30 inches in length. Jacks of 10-ton capacity shall be in accordance with requirements for 5-ton jacks, except they shall have a double socket, and be provided with a 60-inch minimum length lever.

3.11.2.1 Base and frame. Base and frame shall be of cast iron, malleable iron, steel, or aluminum alloy.

3.11.2.2 Rack bar and pawls. Rack bar shall be of steel, properly hardened for the service to be performed. The teeth of the rack bar shall be machine cut and smoothly finished. The lifting and retaining pawls shall be of steel, properly heat treated, and so arranged in the jack mechanism that one or the other will always be in mesh with the rack bar when the jack is under load.

3.11.2.3 Wood base. The 5-ton jack shall be securely fastened to a wood base of sufficient size to provide a firm foundation for the jack. The 10-ton jack shall be securely fastened to an adequate-size, T-shaped wood base. The base shall be constructed of laminated edge-grain hardwood conforming to MIL–L–736, and treated for protection against moisture. When the 10-ton jacks are furnished in pairs, jack bases shall be of right and left hand design to permit two operators to lift and lower cable reels simultaneously.

3.11.2.4 Braces. Each 10-ton jack shall be equipped with three steel braces of ⅞-inch nominal diameter, and of adequate length to provide sturdy support for the jack. Braces shall be securely attached to the jack body near the top, and to the wood base for stability and reinforcement.

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3.11.2.5 Jacks shall conform to table VII for the capacity specified (see 6.2), and be similar to figure 7.

Table VII. Type III. style A. rack bar jacks.

Rated capacity	* Closed height (max.)	Safe rise of screw (min.)	Spindle capacity of cap diameter (min.)	* Lifting hooks (3-level Range)	Reel diameters jacks will handle		Weight of each jack (max.)
					Low rise	Raised height	
Tons	Inches	Inches	Inches	Inches	Inches	Inches	Pounds
5	21	10	2½	9½, 15½, 21	20	60	60
10	30	14	3	17, 23, 29	35	85	120

* Measured from center of 2-inch diameter axle resting in U opening of lifting cap, or hook and includes the wood base.

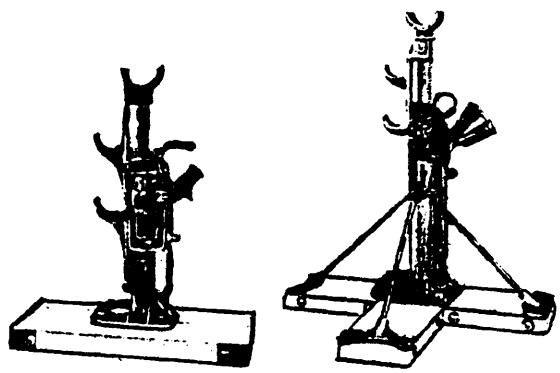


FIGURE 7.—Type III, style A, rack bar jack.

3.11.3 Style B. screw jacks of 5- and 20-ton capacities. Jacks of 5- and 20-ton rated capacities shall have rectangular bases with frames of welded steel. or of securely bolted and braced construction. The jacks shall be operated by inserting a lever bar in either a 3-way. or 4-way nut on the jack screw, and turning the nut.

3.11.3.1 Screw. The screw for style B jacks shall be of one-piece steel and threaded not less than 20¾ inches in length from the bottom. The thread shall be National Acme. 2 inches major diameter and 1¾ inches minor diameter in accordance with 3.2.6. The upper end of the screw shall have a U-shaped swivel or saddle cap designed to safely support a fullly loaded reel.

3.11.3.2 Saddle cap. The saddle cap shall be removable for normal maintenance and shall be designed to rotate freely. Lateral movement of the saddle cap shall not exceed ⅛ inch.

3.11.3.3 Operating lever. The lever shall be of steel. a minimum of 40 inches long, and 1⅛ inches nominal diameter. It shall fit into the nut. or socket of the ratchet mechanism. and be capable of raising the loads specified herein without damage to the jack or lever.

3.11.4 Style B. 15-ton capacity. Jacks of 15-ton rated capacity shall consist essentially of a body. screw. ratchet mechanism. saddle cap, operating lever. wood base. and braces.

3.11.4.1 Body. The body shall be of cast iron. malleable iron. or steel. threaded internally from the top downward for a distance of not less than 3 inches and perpendicular to the base. The thread shall be National Acme. 2 inches major diameter and 1¾ inches minor diameter in accordance with

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3.2.6. The base of the body shall be rectangular. The minimum overall length of the body shall be 18½ inches and the minimum length, width, and thickness of the base of the body shall be 9½, 6, and ½ inches, respectively.

3.11.4.2 Screw. Screw shall be in accordance with 3.11.3.1.

3.11.4.3 Ratchet mechanism. The ratchet mechanism shall be securely attached to the jack screw immediately below the saddle cap but shall be removable for normal maintenance. The gear and pawl shall be of steel and the enclosure (body) shall be of malleable iron or steel. Means shall be provided for reversing the mechanism readily without the use of special tools. The mechanism shall be incapable of changing direction, except when reversed by the operator. The enclosure (body) shall have a socket to properly receive the operating lever.

3.11.4.4 Saddle cap. Saddle cap shall be in accordance with 3.11.3.2.

3.11.4.5 Operating lever. Operating lever shall be in accordance with 3.11.3.3.

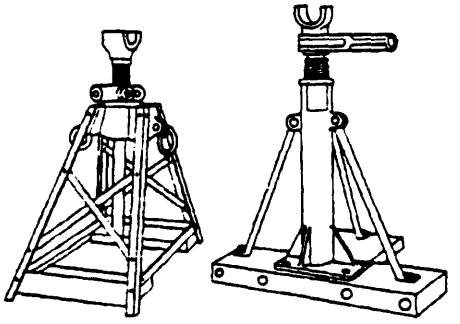
3.11.4.6 Wood base and braces. The wood base shall be constructed of laminated edge-grain hardwood conforming to MIL–L–736, and treated for protection against moisture. The laminations shall be approximately 2 inches wide by 2½ inches thick and tightly bolted together to form a T-shape base. The base shall be approximately 23½ inches across the top of the T, 16 inches from the bottom of the leg to the top of the T, and 8 inches wide across each section of the T. Each jack shall be equipped with three steel braces of ⅞-inch nominal diameter and 17½-inches minimum length. They shall be securely attached to the jack body near the top and to the wood base for stability and reinforcement.

3.11.5 Style B jacks shall conform to table VIII for the capacity specified (see 6.2), and be similar to figure 8.

Table VIII. Type III, style B screw jacks.

Rated capacity	* Closed height (maximum)	Safe rise of screw (minimum)	Spindle capacity of the cap diameter (minimum)	Reel diameters that jacks will handle		Diameter of screw (minimum)	Size of base (minimum)	Weight of each jack (maximum)
				Low height	Raised height			
Tons	Inches	Inches	Inches	Inches	Inches	Inches	Square inches	Pounds
5	42	18	3½	90	114	2	920	125
15	26	17	2¼	50	84	2	240	80
20	28	16	2	52	84	2	320	80

* Measured from center of 2-inch diameter axle resting in U opening of lifting cap and includes the wood base.



Design A.—5 and 20 ton. Design B.—15 ton.

Nonratcheting Ratcheting

FIGURE 8.—Type III, style B, screw jacks.

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3.12 Type V, planer (leveling) jacks. Type V jacks shall be the conventional class of planer jacks composed essentially of a barrel, screw, and swivel cap.

3.12.1 Barrel. The barrel shall be of iron or steel properly threaded to receive the screw. It shall be provided with a large bearing surface at the base, the latter to be finished or cast perpendicular to the axis of the screw. The upper portion of the barrel shall be provided with a positive clamping or locking device, so that the screw may be safely secured at any desired height within its rise.

3.12.2 Screw. The screw shall be of steel, and threaded to properly engage the threaded portion of the barrel. The top portion of the screw shall be of spherical shape and shall be provided with a steel or malleable iron swivel (socket-type) cap, having a flat top surface. The flat top surface of the cap shall be adjustable in any direction, to an angle of not less than 20° with the center of the main screw, or a total of a 40° included angle. The upper portion of the screw shall also be provided with a hexagon section so that it may be raised or lowered with a wrench.

3.12.3 Type V jacks shall conform to table IX for the capacity specified (see 6.2), and be similar to figure 9.

Table IX. Type V, planer (leveling) jacks.

Rated capacity	Closed height (maximum)	Rise (minimum)	Weight of jack (minimum)	Size of main screw (minimum) (diameter)	Size of cap (diameter) (minimum)	Size of base (minimum)
Tons	Inches	Inches	Pounds	Inches	Inches	Square inches
2	2⅞	1	1½	¾	1¼	3.9
4	3¾	1½	3	1	1¾	7
6	5¼	2¼	6	1¼	2	12
8	7½	4½	12	1½	2½	20

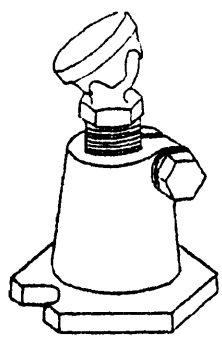


FIGURE 9.—Type V, planer (leveling) jack.

3.13 Type VI, rack bar, (ratchet-lever lowering, lever type). Type VI jacks shall consist essentially of a base and frame, a rack bar, retaining and lifting pawls, trunnions, bearings, socket lever, lever bar, and cap. The jacks shall be single acting, and operate on the downward stroke of the lever.

3.13.1 Type VI, class 1, solid base with foot lift. The base and frame of type VI, class 1 jacks shall be of one-piece construction and of malleable iron, steel, or aluminum. The rack bar shall be solid steel, properly hardened and equipped with a forged claw or foot lift. The teeth of the rack bar shall be machine cut and smoothly finished. A suitable device shall be provided to permit the jack, while under load, to be lowered safely, when and as controlled by the operator.

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3.13.1.1 Pawls. The lifting and sustaining pawls shall be of steel, properly heat-treated and so arranged in the jack mechanism that one or the other will always be in mesh with the rack bar when the jack is under load.

3.13.1.2 Trunnions and bearings. The trunnions, on which the lever socket oscillates, shall be of steel and the bearings, in which the trunnions rotate, shall have case-hardened bushings.

3.13.1.3 Carrying handles. Handles shall be provided on all jacks. Suitable drop-type handles, where practical, shall be provided on jacks weighing in excess of 60 pounds.

3.13.1.4 Type VI, class 1, style A, single socket. Type VI, class 1, style A jacks of 5-, 10-, and 20-ton rated capacities shall have a round single socket and a 1¼-inch minimum diameter steel lever bar for operating the jack, except that for the 5-ton, a 1-inch diameter steel lever bar will be acceptable. Style A jacks of 15-ton rated capacities shall have a larger single socket.

3.13.1.5 Class 1, style A, single-socket jacks shall conform to table X for the capacity and height specified (see 6.2), and be similar to figure 10.

Table X. Type VI, class 1, styles A and B rack bar (ratchet-lever lowering, lever type), solid base with foot lift, single- or double-socket jacks.

Rated capacity	Closed height (maximum)	Rise (minimum)	Weight of jack (maximum)	Size of base (bottom) (minimum)	Head (cap) size (minimum)
Tons	Inches	Inches	Pounds	Square inches	Square inches
5	17	9	45	36	5½
10	22	11½	65	54	7½
15	23¼	11	93	75	9½
15	28¼	17	100	75	9½
15	35	23½	125	75	9½
20	23¼	17½	110	85	9½

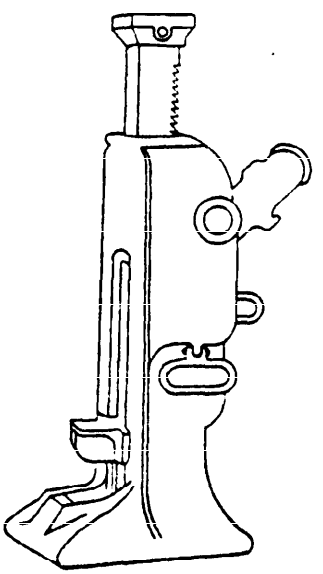


FIGURE 10.—Type VI, class 1, style A, rack bar (ratchet-lever lowering, lever type), solid base with foot lift, single-socket jack.

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3.13.1.6 Type VI. class 1. style B. double socket. Type VI. class 1. style B jacks shall be in accordance with class 1. style A jacks except all capacities shall have round double sockets and a 1¼-inch minimum diameter steel lever bar. except that for the 5-ton, a 1-inch diameter steel lever bar will be acceptable. Jacks shall conform to table X for the capacity and height specified (see 6.2), and be similar to figure 11.

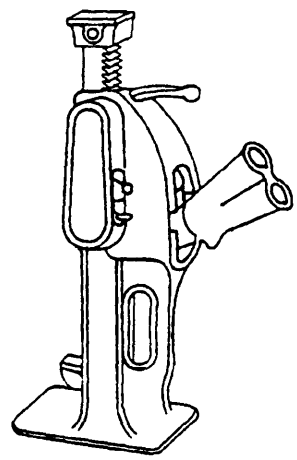


FIGURE 11.—Type VI. class 1. style B. rack bar (ratchet-lever lowering, lever type), solid base with foot lift, double-socket jack.

3.13.2 Type VI. class 2. hinged base. double socket. Type VI. class 2 jacks shall consist essentially of a base and frame. rack bar. retaining and lifting pawls. trunnions. bearings. socket lever, lever bar. chain and cap. The jacks shall be single acting, and operate on the downward stroke of the lever.

3.13.2.1 Base and frame. The base shall be suitably hinged to the frame so as to permit the frame to swivel away from the operator at an angle of 30° (plus or minus 5°) to a perpendicular centerline of the base. Unless otherwise specified (see 3.13.2.5), the base and frame shall be of malleable iron or steel.

3.13.2.2 Socket and lever bar. The jacks shall have round double sockets and a 1¼-inch diameter steel lever bar approximately 5 feet long.

3.13.2.3 Carrying handles. Suitable handles (drop type where practical) shall be provided on all jacks.

3.13.2.4 Type VI. class 2. style A. with foot lift. Unless otherwise specified (see 6.2), type VI. class 2. style A jacks shall be furnished with a 5-foot minimum length of ½-inch chain in accordance with type I. grade C. class 3 of RR–C–271. A grab hook in accordance with type V. class 2 of RR–C–271 shall be included on one end of the chain. A lifting hook may engage a recessed cap, or be attached to the other end of the chain. similar to that shown on figure 12. A pear-shaped link may be substituted for the lifting hook attached to the chain. The lifting hook or link shall be of comparable strength to that of the chain in order to provide efficient performance.

3.13.2.4.1 Rack bar with foot lift. Style A jacks shall have a solid rack bar of steel, properly hardened and equipped with a forged claw or foot lift. The teeth of the rack bar shall be machine cut and smoothly finished. A suitable device shall be provided to permit the jack. while under load. to be lowered safely when and as controlled by the operator.

3.13.2.4.2 Pawls. The lifting and sustaining pawls shall be of steel. properly heat treated. and so arranged in the jack mechanism that one or the other will always be in mesh with the rack bar when the jack is under load.

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3.13.2.4.3 Trunnions and bearings. The trunnions on which the lever socket oscillates shall be of steel and the bearings in which the trunnions rotate shall have case-hardened bushings.

3.13.2.4.4 Cap. The cap shall be slotted or recessed so as to suitably engage any link of the chain.

3.13.2.4.5 Base-locking hook. Style A jacks shall be provided with a positive-locking hook to eliminate swivel action so that they may be used as a solid-base jack.

3.13.2.4.6 Type VI, class 2, style A jacks shall conform to table XI and be similar to figure 12.

Table XI. Type VI, class 2, style A, rack bar (ratchet-lever lowering), hinged base, double-socket jack with foot lift.

Rated capacity	Closed height (maximum)	Rise (minimum)	Weight of jack complete (maximum)	Size of base (bottom) (minimum)	Head (cap) size (minimum)
Tons	Inches	Inches	Pounds	Square inches	Inches
15	23¼	12¼	112	75	3 x 3

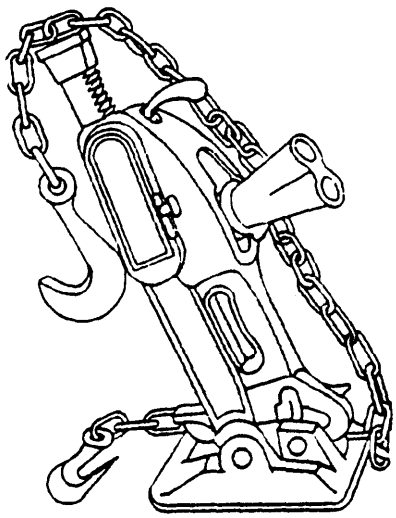


FIGURE 12.—Type VI, class 2, style A, rack bar (ratchet-lever lowering), hinged base, double-socket jack with foot lift.

3.13.2.5 Type VI, class 2, style B, pole pulling. Type VI, class 2, style B jacks shall be designed for pulling or straightening the larger sizes of telephone, telegraph, and power poles. They shall be in accordance with class 2, style A jacks, except they shall not have a foot lift, and the sub-base shall be constructed of 10-inch standard steel I-beam, or a 10-inch channel weldment of equal weight and capacity. Jacks shall be furnished with a 6-foot minimum length of ⅝- or ¾-inch chain in accordance with type I, grade C, class 3 of RR–C–271, with a suitable pear-shaped link included on one end. When specified in the contract or order (see 6.2), an aluminum alloy frame or housing shall be provided.

3.13.2.5.1 Type VI, class 2, style B jacks shall conform to table XII and be similar to figure 13.

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Table XII. Type VI, class 2, style B, rack bar, (ratchet-lever lowering), hinged-base, double-socket, pole pulling jack.

Rated capacity	Closed height (maximum)	Rise (minimum)	Weight of jack complete (maximum)	Size of base (minimum)
Tons	Inches	Inches	Pounds	Square inches
15	38¼	22	250	200

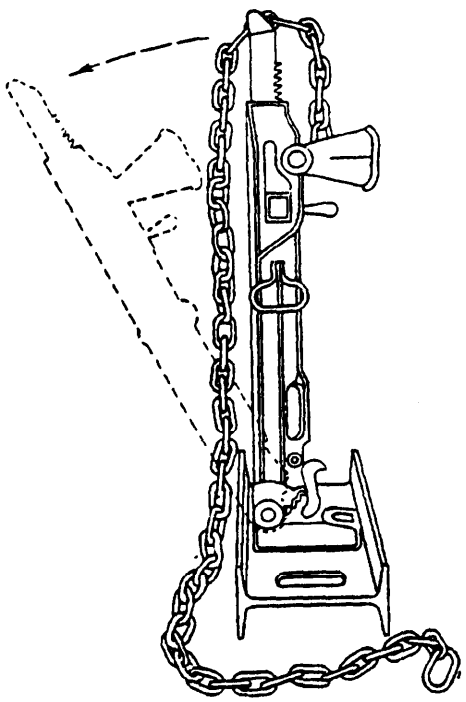


FIGURE 13.—Type VI, class 2, style B, rack bar (ratchet-lever lowering), hinged base, double-socket, pole-pulling jack.

3.14 Workmanship. The jacks shall be free from defects that may impair their use under normal service conditions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Inspection of materials and components. In accordance with 4.1. the supplier is responsible for insuring that materials and components used were manufactured. tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein. or. if none. in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL–STD–105. Data for sampling shall be as stated in table XIII.

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design. material. finish. coating. construction. workmanship. and marking.

4.3.2 Dimensional and weight examination. Each sample unit shall be examined for any nonconformance with dimensional and weight requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1 through 4.4.3.

Table XIII. Sampling data.

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	One ea.	II	1.5	Defects per 100 units	4.3.1
Dimensional and weight examination	do.	II	1.5	do.	4.3.2
Testing	do.	S–4	4.0	do.	4.4.1. 4.4.2. 4.4.3
Preparation for delivery	One container	S–2	4.0	do.	4.5

4.4.1 Performance test procedures. Each of the sample jacks shall be subjected to the load tests specified in 4.4.2 and 4.4.3. Jacks and their component members shall show no mechanical defects or ineffective operation and shall not fail. fracture. take permanent set. or show any evidence of instability. bending. shearing. or undue or excessive wear. Jacks and their component control devices and trip mechanism shall operate properly and effectively. and there shall be no mechanical defects at the completion of tests.

4.4.2 Load tests. Each sample jack shall support the respective test loads specified in table XIV. The jack shall be in a vertical position and the load shall be balanced. The load shall be applied at any position designated by the inspector within the specified rise of the jack. except for the following jack:

(a) For type II. the full load shall not be applied before the jack has been raised to at least 50% of its maximum overall height.

4.4.3 Load-sustaining tests. Each sample jack shall sustain (support) for 10 minutes the respective test loads specified in table XIV. The load shall be applied at maximum rise.

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Table XIV. Load tests.

Type	Jack	Rated capacity	Test-load support		Test sustain load for 10 minutes
			Head, saddle (cap), hook lift	Foot lift	Head, saddle (cap), hook lift
		Tons	Tons	Tons	Tons
I	Screw, hand:				
	Class 1, style A, standard speed, upright or inverted with foot lift	15	16½	7½	18
		25	27½	12½	30
		35	37½	18	40
		50	55	25	59
		75	83	38	85
	Class 1, style B, standard speed, journal or inverted	15	16½	--------	18
		25	27½	--------	30
		35	38	--------	40
		50	55	--------	58
	Class 2, high-speed lever- or governor-lowering control	25	27½	--------	30
		35	38	--------	40
		50	55	--------	58
		75	83	--------	85
		100	110	--------	115
	Class 3, bell bottom	5	6	--------	6½
		8	9	--------	10
		12	13	--------	15
		15	17	--------	18
	Class 4, geared screw (for auto or truck)	1½	1¾	--------	2
		2	2¼	--------	2½
		3	3¼	--------	3¾
		5	5½	--------	6
II	Scissors, hand	1½	1¾	--------	2
		2	2¼	--------	2½
III	Reel, hand (cable, wire, and wire rope):				
	Style A, rack bar	5	5½	--------	6
		10	11	--------	12
	Style B, screw	5	5½	--------	6
		15	16½	--------	18
		20	22	--------	22
V	Planer (leveling)	2	--------	--------	2½
		4	--------	--------	5
		6	--------	--------	7¼
		8	--------	--------	10

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Table XIV. Load tests (cont'd).

Type	Jack	Rated capacity	Test-load support		Test sustain load for 10 minutes
			Head, saddle (cap), hook lift	Foot lift	Head, saddle (cap), hook lift
		Tons	Tons	Tons	Tons
VI	Rack bar (ratchet-lever lowering)				
	Class 1, style A and B, solid base with foot lift, single and double socket	5	5½	2½	5½
		10	11	5	11
		15	16½	7½	16½
		20	22	10	22
	Class 2, style B, hinged base, double socket, pole pulling	15	16½	7½	16½
		15	16½	--------	16½

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table XV. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL–STD–105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery.

Table XV. Classification of preparation for delivery.

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size; location; sequence; or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified (see 6.2), the jacks shall be preserved, packaged, and packed in accordance with MIL–H–15424. The level of preservation and packaging shall be A or C and the level of packing A, B, or C as specified (see 6.2).

5.1.1 As part of the requirements of section 5 of this specification, table XVI is considered as part of and used in lieu of that portion of table I of MIL–H–15424 covering jacks.

Table XVI.

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7
Jack:						
Mechanical, hand						
Operated						
Screw, hand						
up through 22 pounds	P–2, P–7	I	1		6	6
23 pounds to 35 pounds	P–2, P–7	I	1		4	4
36 pounds to 85 pounds	P–2, P–7	I	1		2	2
85 pounds and over	P–2, P–7	I			1	1
Scissors, hand	P–2, P–7	I	1		12	12
Reel, hand, cable, wire, and wire rope	P–2, P–7	I			1	
Planer, leveling						
up through 6 pounds	P–2, P–7	I	1		12	12
7 pounds to 12 pounds	P–2, P–7	I	1		10	10
13 pounds and over	P–2, P–7	1	1		6	6
Rack bar, ratchet-lever lowering	P–2, P–7	I			1	1

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military activities. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Intended use.

6.1.1 Type I, class 1, style A jacks are for heavy duty use in handling heavy machinery, construction jobs, and rigging. Parts are enclosed for protection from dirt or weather in outside service.

6.1.2 Type I, class 1, style B jacks are for heavy duty use, as are style A jacks except they are adaptable in space-restricted areas such as for inspecting and renewing journal brasses on railroad trains.

6.1.3 Type I, class 2 jacks are for heavy duty use, such as handling loaded freight cars or locomotives, and are of high speed, self-lowering control.

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6.1.4 Type I, class 3 jacks are for use in construction, maintenance, rigging, and for general purposes.

6.1.5 Type I, class 4 jacks are for use on automobiles and lightweight trucks.

6.1.6 Type II jacks are for use on automobiles and light trucks.

6.1.7 Type III jacks are for use in handling heavy reels of cable, wire, wire rope, etc.

6.1.8 Type V jacks are for use in leveling and for support when clamping down castings, forgings, and weldments preparatory to machining operations. Also for setting up material on large surface plates for layout and inspection.

6.1.9 Type VI, class 1, styles A and B jacks are for use in construction, shoring, rigging, moving machinery, and for general industrial use.

6.1.10 Type VI, class 2, style A jacks are for use by riggers, in maintenance, construction, moving machinery, and for general all-purpose jacking.

6.1.11 Type VI, class 2, style B jacks are for use in pulling or straightening all sizes of telephone, telegraph, power, and other poles.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following data in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of jack, class, and style, as applicable (see 1.2).
(c) Rated capacity (see applicable table).
(d) When identification marking with decals is not acceptable (see 3.5).
(e) When acrylic coating on decals is required (see 3.5).
(f) Additional equipment as required (see 3.6).
(g) Finish, if other than manufacturer's standard is required (see 3.7).
(h) When steel axle is required (see 3.11.1).
(i) If chain is not required, so specify (see 3.13.2.4).
(j) When aluminum alloy frame or housing is required for type VI, class 2, style B jacks (see 3.13.2.5).
(k) Selection of applicable level of preservation, packaging, and packing required (see 5.1).

6.3 For civil agency procurements. When level B packaging is required for civil agency procurements, the level A requirements in 5.1 shall apply.

6.4 Supersession data. Jacks formerly included in GGG–J–51c, but not presently shown in GGG–J–51e, are now covered as follows:

GGG–J–51c	GGG–J–51e
For all hydraulic jacks see GGG–J–63.	
Type IV.—Dolly type, mechanical.	Deleted.

MILITARY INTEREST:	Review activities	User activities
Custodians	MO	CE, WC, EL
Army—GL	SH	YD, MC, WP
Navy—SH	Waived coordination	
Air Force—69		
Preparing activity		
GSA–FSS		

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